

Def. A subnormal series of a group G is a finite sequence $\{H_i\}$
 $\{e\} = H_0 \triangleleft H_1 \triangleleft H_2 \triangleleft \dots \triangleleft H_n = G.$

Def. A composition series of a group G is a subnormal series $\{H_i\}$
 $\{e\} = H_0 \triangleleft H_1 \triangleleft \dots \triangleleft H_n = G$ 이고 모든
 $i = 0, 1, \dots, n-1$ 에 대해 H_{i+1}/H_i 가 단순군이다.
즉, subnormal series를 얼마나 더 잘게 쪼갤 수
있다.

Theorem.1 Jordan-Hölder
Any two composition series of G are
isomorphic.

증명). $\{H_i\}, \{K_j\}$ 가 G 의 composition series
라고 하자. $\{H_i\}, \{K_j\}$ 는 서로 isomorphic 한
refinement를 가진다. 그런데 $\{H_i\}, \{K_j\}$ 는 더
이상 refinement를 진행할 수 없으므로
 $\Rightarrow \{H_i\} \cong \{K_j\}.$

Theorem 2.

If G has a composition series and N proper normal subgroup of G , then G has a composition series containing N

증명.

$\{e\} \triangleleft N \triangleleft G \rightarrow$ subnormal series.

사이 사이에 subnormal group $\frac{e}{2} \frac{e}{2}$ 넣는다.

Theorem 3.

If G is a finite group, G has a composition series.

증명.

i) $|G|=1 \Rightarrow G = \{e\} \Rightarrow H_0 = \{e\}$ is a composition series.

ii) $n \in \mathbb{N}$, G has a composition series if $|G| \leq n$.
Assume $|G|=n+1$.

Case 1. G is simple. $\Rightarrow \{e\} \triangleleft G$ is a composition series

Case 2. G is not simple.

$$\exists N \triangleleft G \quad (N \neq G, N \neq \{e\})$$

$$\{e\} \triangleleft N \triangleleft G$$

$|N| < |G| = n+1 \Rightarrow |N| \leq n \Rightarrow N$ has a composition series by Mathematical Induction.

$$\{e\} = N_0 \triangleleft N_1 \triangleleft N_2 \triangleleft \dots \triangleleft N_m = N \rightarrow \text{composition series.}$$

$\rightarrow N_{i+1}/N_i \cong \mathbb{Z}_2$.

$|G/N| \leq n \Rightarrow G/N$ has a composition series. $\therefore$ G/N is simple.

$$\{e\} = M_0 \triangleleft M_1 \triangleleft M_2 \triangleleft \dots \triangleleft M_k = G/N \quad \text{composition series.}$$

$$\begin{aligned} \gamma: G &\rightarrow G/N \\ g &\mapsto gN. \end{aligned}$$

$$N = \gamma^{-1}(M_0) \triangleleft \gamma^{-1}(M_1) \triangleleft \dots \triangleleft \gamma^{-1}(M_k) = G.$$

$$\hookrightarrow \gamma^{-1}(M_{j+1}) / \gamma^{-1}(M_j) \cong M_{j+1} / M_j \text{ 인 걸 보인다!}$$

반대로, Z 는 composition series 를 갖지 않는다.

Z 의 비자명한 subgroup 들은 nZ 형태.

$$H_{i+1}/H_i \cong nZ \cong Z \rightarrow Z \text{ 는 simple group 이 아니다,}$$

Z 는 composition series 를 갖지 않는다.

Def. A group G is solvable if it has a composition series $\{H_i\}$ such that all factor group H_{i+1}/H_i are Abelian.

✗ $n \geq 5$ 일 때, S_n 은 solvable 이 아니다.

$\{e\} \triangleleft A_n \triangleleft S_n$ 인데, $A_n/\{e\}$, S_n/A_n 어느 것도 Abelian 이 아니다.

A_5 가 가장 작은 non-solvable group.

↳ $|A_5| = 60$ 보다 작은 위수에서는 solvable.